

Design for Social Good

A curious dive into designing for social good in an attempt to solve one of India's wicked problems - female literacy, both children and adults. This was an optional coursework assignment that I worked on and felt strongly about. I thought it would be a good opportunity to apply a design thinking approach to solving something as critical and complex as literacy in a developing country.

My intention was to explore the problem together with the intricacies that fuel it, illuminating readers about its complexity, while reimagining a solution that makes a dent in it. I believe that technological progress, particularly artificial intelligence, will play a huge role in bringing us closer and quicker to a utopian state.

Solving this issue requires the country to come together, to elevate itself and eliminate the burden of illiteracy and unemployment which exhibit further downstream concerns. While these perils are severely exploited for selfish gain, it is imperative that the nation addresses them to secure a brighter future.

TL;DR : [View Comparison Report](#)

This comparison report is based on a simple web crawl using AI and may not be 100% factually accurate.

Improve literacy and employment rates for women in India

Introduction |

India is home to nearly 20% of the planet's human population. Over 1.4 billion (2025) people co-exist in the seventh largest country by

land area. With a GDP of over \$4.4 Trillion (2025), India has established itself as the 5th largest economy in the world.

Don't let those numbers fool you. A large population produces its fair share of difficulties. It creates a massive strain on access to and availability of drinking water, food, sanitation, healthcare, employment, infrastructure, public services, natural resources, and more.

Obviously, educating a populace of this size is an enormous and complex task. According to the National Statistical Commission (2018), a whopping 20% or 280 million Indians are uneducated. Disparities exist between urban and rural literacy rates at the national level - 90% and 77.5% respectively (2023). This differs significantly by state as well.

Further, India suffers from gender imbalance. Males have a higher literacy rate (87.2%) than females with a 12.5% gap. Rural and state dependencies impact this gap further. This imbalance permeates deeply into other facets of everyday life too.

Government initiatives like the Right to Education Act, NGOs, and Non-profits have helped improve enrolment rates for females in recent times. In fact, free education at the primary level has shown remarkable results. Unfortunately, drop-out rates at later stages of education are equally alarming. However, a primary education alone does not guarantee a better outcome in employment or income.

Vocational training centres have attempted to bridge skill gaps for advanced ages. While critically dependent on access and adoption, it hasn't produced the intended results due to gender disparities, financial issues, technology access, social norms, and domestic commitments of women. Worse, post-vocational training, employers prefer not to hire these women in spite of being skilled for many of the reasons listed above.

Having surfaced these issues, it is well acknowledged that literacy dispels ignorance and dependence while propelling citizens to become productive members of society and the economy. So, how might we

improve the literacy and subsequently employment rates of women in India?

New India Literacy Programme (NILP), Saakshar Bharat Mission, Sarva Shiksha Abhiyan (SSB), Rashtriya Madhyamik Shiksha Abhiyan (RMSA)

National Schemes I

As per Indian census standards, a person aged seven or older is considered literate if they can both read and write with understanding in any language. Recently, the Ministry of Education (MoE) expanded this to include the ability to compute with comprehension, identify, understand, and create information, and possess critical life skills like digital and financial literacy.

[Resource+](#)

Definition of Literacy I

Dependency on volunteerism, lack of appropriate incentives - learners and volunteers, lack of localisation, no guarantee of lifelong learning, weak retention mechanisms, secure government employment, low target metrics, no post certification employment certainty, inadequate awareness campaigns, lack of a systemic approach, lack of system audits, etc.

The target of make 5 crore learners (age 15+) literate between 2022-2027 is not conducive to solving the evolving population and illiteracy growth rates.

Drawbacks I

Building empathy

Desk Research I

Secondary data

-
- Web Sources

Field research I

Interviews & Surveys

- Interviews

- Surveys

Learnings I

Responses from Field Research

The responses can be distilled into categories. They are by no means comprehensive, but serve as illuminating examples of the critical issues :

Family, Social and Economic Constraints

- Migration of working parents to new locations that presented employment opportunities
- Forced to assist parents in their occupational work
- Sent to perform menial work for extra income
- Domestic responsibilities due to working parents
- Caring for elderly members in parents absence

- • Restricted movement for girls outside their homes or village
- • Lack of perceivable impact on income in the short-run
- • Inability to afford transportation, tuition or textbooks
- • Child marriage
- • Early pregnancies
- • Poor health and well-being

Infrastructure

- • Distant proximity to educational facilities
- • Lack of adequate transportation
- • Lack of decent seating in schools
- • Poor classroom accommodation
- • Frequent power cuts
- • Poor or no availability of technological support or access
- • Poor or no internet connectivity
- • Lack of educational materials to serve attendees
- • Poor sanitation and unhygienic conditions
- • Low availability of clean drinking water
- • Lack of meal plans for attendees

Staff & Government Involvement

- • A one-size fits all strategy
- • Lack of adjustments of the local social, economic, and cultural influences on the ground
- • Irregular attendance of teachers and support staff

- • Lack of appropriate training and up-skilling of teachers and staff
- • Outdated curricula
- • No real strategies for retention and drop-out prevention
- • Lack of financial backing to maintain or expand facilities
- • Corruption
- • Poor or no coordination
- • Lack of course correction to address outcomes

Communication

- • Lack of awareness around education programmes
- • No knowledge of help/contact centres for information
- • Disconnected action plans and poor outreach programmes to advise the local population
- • Inability to communicate the link between education and economic prosperity
- • Limited or no access to technology inhibits reach

Other Reasons

- • Insecurity of having girls travel to remote locations
- • Families see expense for girls education as a liability especially when they are already at or below the poverty line
- • Education is not a priority for women of advanced age
- • Social, community or peer pressure

- No on-site accommodation, like hostels
- Poor performance in class due to need for personal attention

Persona types

Defining the problem

A Wicked Problem I

India is a complex country with 28 states and 8 union territories, each with its own unique story, culture, heritage, social norms, religious denominations, festivals, natural resources, demographics, power dynamics, idiosyncrasies, and more. Tolerance and unity are not unanimously shared.

India is deep-rooted in tradition, customs, superstitions, and formality. Class and caste distinctions are still practiced. Gender inequality is a major issue. There is a strong preference for males who enjoy more autonomy than females. Social and communal ideologies benefit males over females.

Many females are subjected to the perils of child marriage, teenage pregnancy, domestic work, poor nutrition, exploitation, dowry, income discrimination, violence, etc. In addition, they are not permitted access to education in the same way as males.

Corporates prefer not to hire women for fear of attrition, restricted work hours, maternity regulations, protection policies, and demands in their family life.

The Indian government spends only 4.1% of its gross domestic product (GDP) on education compared to 3.8% by China, the world's

second most populated country. What is important here, is that China's GDP is 4.1 times that of India. While the population sizes are similar, the educational budgets differ significantly.

In addition, the country suffers from widespread corruption, polarizing political ideologies, religious differences, infrastructural problems, inflation, joint-family systems, notable income and quality of life disparities, and more.

The interconnected nature of the many variables described above indicate a non-linear path to a solution. Problems like these cannot be defined simply and solving them isn't straightforward.

Young girls, adolescents, and women have been subjected to social and economic discrimination for decades. Education is instrumental in addressing these issues while helping females achieve their full potential, and improving their quality of life.

However, its acceptance as a means to progress has been disproportionately adopted for various reasons that I've sighted earlier. Hence, females have not been able to utilize educational programmes to improve their circumstances in society.

A lack of education increases female dependency on immediate family, in-laws, or male counterparts and very often limits their employment opportunities.

The issues begin in early childhood and continue into adulthood triggering a cyclical effect for subsequent generations.

Problem Statement I

If we create a modular strategy to tackle issues at the local level while linking their implementation to state, regional, and central agencies along with monitoring, auditing, and support, we could lead a coordinated effort in bringing about profound change.

By joining hands, sharing information, learnings, best practices, and holding the system accountable, we could improve enrolment rates for educational programmes aimed at females of all age groups and ultimately their integration into the economy as productive members

of society.

Hypothesis Statement I

Education is a liberator in many ways. It helps reduce the impact of poverty, has a positive influence on health, increases participation in the labour market, and improves financial security.

It promotes empowerment, confidence, independence, personal development, and elevates the overall quality of life for people.

Impact I

Ideation

Mind-Mapping I

As mentioned earlier, a non-linear path to a solution seemed inevitable. Initially, I tried different ideation techniques, ultimately settling on mind-mapping as it captured the convoluted essence of the problem.

I kept literacy as the central idea. Although literacy isn't imperative to employment, the quality of employment types is dependent on literacy levels. Therefore, employment types ultimately dictate one's income, health, and overall quality of life.

Other techniques like brainstorming, worst possible idea, reverse brainstorming, how might we, and challenging assumptions were used.

Other Techniques I

L.E.A.D Programme I

With limited access to subject matter experts, I relied on available resources to propose a solution. I settled on the idea of a L.E.A.D programme, which stands for **Literacy** and **Employment** through **Access** and **Development**. Of course, an adaptation of the nomenclature would be necessary.

What is the L.E.A.D programme about and how does it work?

L.E.A.D is a modular approach to addressing literacy in India. Operating across all levels of the country, the programme recognises the need for a coordinated effort in addressing this complex problem. In other words, it is a systemic approach to solving a multi-faceted problem.

The Central government in partnership with an independent international organisation, such as UNESCO, spearhead the programme. Together, they work with state governments, vocational institutions, NGOs, the private sector, and citizens to tackle illiteracy at scale.

Stakeholders to the mission form a value-chain. Each entity is responsible for and accountable to the mission's goal. In collaboration, they study situations as they filter through to the target locations - centre, region, state, city, district, village, etc. This creates a tree-like structure where the central government and independent agency function as the trunk.

Teams are adequately staffed and use a single digital collaboration tool to consolidate, communicate, and monitor the process. This means, even at the grass roots, simple messaging modes like 'SMS' can be used to drive information upwards.

Research, information, best practices, case studies, and more are actively shared and leveraged throughout the value-chain and across networks of value-chains. The process is audited to ensure conformity by members and who are incentivised for their contribution.

L.E.A.D also runs support groups in parallel to help learners manage and deal with socio-cultural pressures to ensure learner retention in the programme. This includes conversations and awareness campaigns with family, prominent figures, and other members in the

location of the learner to gain acceptance.

Audited reports of programme implementations are circulated to stakeholders for deliberation and if needed, course correction. They are also periodically published online for public perusal.

The programme focuses on action and outcomes, ensuring accountability through Objectives & Key Results (OKRs). It is based on the pillars of **adaptation**, **inspection**, **collaboration**, and **transparency**, which are critical to its success.

Overview I

L.E.A.D focuses on building:

- Foundational literacy, numeracy, and conversational confidence
- Skills in digital awareness and interaction
- Critical life skills in hygiene, healthcare, civic sense, and financial literacy.
- Knowledge around family planning and the importance of education in economic improvement.
- Vocational skills relevant to industry, particularly in proximity to the learner's location.
- Firm understanding and necessity of constant learning to raise economic status and skill relevancy.
- Support groups and networks, not only for learners, but also to help learners continue their learning journey by mediating domestic conflicts and pressures.

Stakeholders

Who's Involved I

International Organisation/s can help provide financial assistance,

train teachers and administrators in promoting literacy, monitor and allocate resources to high impact channels, evaluate and recommend curricula, advise and advocate literacy policies, build partnerships between stakeholders, conduct research, share best practices etc.

The **Central Government** orchestrates the execution of literacy campaigns through the arms of state governments, monitors the flow of financial support, audits consumption and outcomes of initiatives, formulates policies, allocates funding, partners with international organisations and countries, builds public and private sector partnerships, invests in technology, recruits teachers, administers training programmes etc.

State Governments can execute the Center's policies and programmes, work with NGOs, the private sector, and social organisations within the state, build infrastructure, monitor programme effectiveness, ensure curriculums are relevant and updated, raise literacy awareness, offer financial assistance, support, train volunteers etc.

NGOs can implement literacy programmes tailored to their localities, collect and distribute education supplies, contribute to educational infrastructure development, encourage parental buy-in, collaborate with state governments, advocate for policy improvement, conduct on-the-ground research, build connections, and share feedback.

The **Private Sector** can contribute employee time to serving the literacy cause, donate computational equipment, furniture, funds, facilities to accommodate learners, develop apprenticeship programmes, scholarships, and partner with NGOs or governments.

Private Schools and Colleges can provide facilities during closed hours, donate textbooks, furniture, technical equipment, school supplies, organise donation drives to collect and distribute student material. Teachers can volunteer time and support learners in areas of expertise. They can partner with NGOs and state governments, contribute through research and feedback, help drive literacy campaigns and foster involvement from the community.

Vocational Institutions can play a big role in up-skilling adults.

They can work with state governments and local industries to fill skill gaps, update and offer relevant courses, encourage entrepreneurship, understand issues on the ground and report back to collaborators to

bring about programme improvements.

Citizens volunteer their time in literacy drives, donate education material, fund initiatives, sponsor students, tutor participants, raise awareness, advocate for better policy etc.

Programme I

Auditors & Localisation Committee

The programme demands system and process audits and hence their appointment by the central government or international organisation to inspect implementation of initiatives on the ground.

Audits can happen at any time and any number of times during the year. They are performed at critical stages where pertinent information, decisions, funding, or incentives are transmitted between entities. This is to ensure strict adherence to programme guidelines.

Besides **auditors**, the **localisation committee** oversees curation of content at the implementation level, ensuring content is adapted to and receptive of those targeted.

Why is L.E.A.D different?

- Integrates and leverages technology at all points in the value chain, both in process and delivery.
- Includes digital literacy within real world contexts, helping learners gain relevant life skills as digital citizens of the 21st century.
- Leverages artificial intelligence (AI) for speed and scale, particularly in complex issues.
- Develops content in a language recipients are comfortable with. Content development is customised locally when necessary. General content would be prepared in a mainstream language.
- Accommodates learning around recipient schedules. Learners are not restricted by time or space. Accessing digital

media ensures learners are not left behind. In addition, local study groups built around common messaging tools like WhatsApp can unite and support learners.

- • Audits, monitors, and reports transparently, regularly and truthfully in the public eye.
- • Incentivises contributions from civilians and the private sector through tax benefits, cost of loans, or discounts on educational courses.
- • Localises curricula to meet current demand while adapting it to changing trends.
- • Leverages a network of agencies in a modular way to address critical issues impacting potential learners.

Recommendations

- • Tackle female illiteracy where it originates through law, and policy with zero tolerance for female education restriction.
- • Advance socio-cultural change and acceptance through a multi-channel strategy - road shows, social media campaigns, cinema etc.
- • Increase teacher headcount through incentivised term-based opportunities for retired teachers.
- • Institute mandatory volunteer hours to increase participation for teachers in training.
- • Offer free/discounted training and up-skill programmes for teachers that volunteer time.
- • Offer tax benefits to rural and scheduled class families supporting female education.
- • Offer welfare schemes in exchange for attendance and certification.
- • Design library programmes for females and foster an

appreciation for reading.

- • Promote reusability and life-cycle extension of digital and computational equipment.
- • Provide lodging and meals during seasonal employment migration for affected children.
- • Encourage and incentivise television, internet, and radio networks to support educational programming.
- • Improve and maintain school infrastructure, including meal plans.
- • Advance socio-cultural change and acceptance through a multi-channel strategy - road shows, social media campaigns, cinema etc.
- • Encourage social interactions focused on vocational courses, assignments, and innovative ideas.
- • Audit and update curricula, consistently aligning it with the foreseeable future.

Communication and awareness

While there are several traditional mediums to advance the goals and objectives of L.E.A.D, I've highlighted a few I consider necessary in its endeavour. The communication strategy should effect positive change by promoting gender equality and advocating for education and employment opportunities for females. By strategically engaging audiences, raising awareness, and fostering a supportive open dialogue, the L.E.A.D programme aims to contribute to the advancement of female rights and their broader inclusion in education and employment.

- • **Social Media:** 67.5% of all internet users in India (regardless of age) access at least 1 social platform. Social media would be pivotal in spreading awareness about female illiteracy and the L.E.A.D programme's work in combating it.
- • **Internet Search:** Internet has a 57% penetration rate

in India. According to [Semrush](#), India accounts for 6% of Google.com website traffic, 87% via mobile. A good SEO and SEM strategy can draw attention to illiteracy and elevate L.E.A.D's role in it.

- • **Television and radio networks:** With a 75% penetration rate, television is largest reach medium. Radio has been growing in tier II and III markets probably due to internet connectivity gaps. This represents a good opportunity to raise awareness and spur discussion.
- • **Streaming media services:** Studies indicate that roughly 61% of streamers are more responsive to advertising on streaming platform, with 48% searching for the product and 33% making a purchase after the fact. Relevant content, messaging, and advertising could help achieve L.E.A.D's goals.
- • **Roadshows:** Roadshows in rural India can help build awareness. Local events are perfect opportunities to engage in dialogue with the local community.
- • **Influencer marketing:** India is highly influenced by celebrity status, particularly uneducated/partially educated segments. While it is counter-productive for some individuals and fraternities to advocate education, there is some scope amongst those who understand the bigger picture and endorse change .

How artificial intelligence (AI) can help

- • AI can help speed up, augment or automate processes, alleviating administrative burden at scale.
- • Assist in the formulation of national, regional, state, and local strategies that cohesively achieve the ultimate objective.
- • Adaptive learning, meeting learners where they are and at a pace they are comfortable with.
- • Voice-enabled learning to address barriers in reading

ability.

- • Generate educational and instructional material curated appropriately for context and target audience.
- • Help develop curricula that are contextually aligned with real world situations and emerging trends.
- • Generate tests, assignments, and projects that offer a practical application of theoretical teachings.
- • Virtual AI tutors that help students dive deeper into subjects or concepts, creating a collaborative learning experience, especially when teachers are not immediately available.
- • Virtual chat assistants to help answer questions or offer guidance via messaging services like WhatsApp.
- • Suggest teaching strategies based on performance data.
- • Generate marketing material, text, visuals, and audio to build awareness campaigns customised to the uniqueness of the variables in play.
- • Cut across national databases to find patterns, spot trends, unearth insights, and identify opportunities for improvement.
- • Run ML models to predict outcomes and take precautionary actions based on historical data, like drop outs.
- • Real-time monitoring of the system at a granular level.
- • Identify lookalike learners to build local study groups or chapters that enhance the learner's experience.
- • Identify hubs, support groups, and other entities in close proximity to target locations to build relationships, consolidate action plans, and deploy consistent strategies around campaigns.
- • Resource and budget allocation to consolidate or

redirect funds appropriately.

Conclusion

L.E.A.D will work with agencies to develop and share impactful stories or documentaries of females who have overcome challenges through education. It will use these narratives to inspire others and advocate for social, cultural, and economic change.

Change will be imperative to the cause. Influencing the ways people think and their ways of life will be challenging. To bring about a paradigm shift, concerted and consistent messaging around the women's rights will be needed to bring about a shift, especially in rural areas.

L.E.A.D will partner with media organisations to create and disseminate compelling narratives that highlight the importance of gender equality, female education, and employment. It will have to do this nationally and locally through a multi-channel approach.

The programme will lobby for workforce rights, ensuring females have an equal right to employment opportunities while also working to advance their presence in male dominated industries.

The programme will need to work with many stakeholders and political parties to foster a sense of pride and patriotism, to inspire people who have the power to act to do so for the betterment of the country. It means that every citizen with the capacity to help, can and should extend their services to the community and beyond. CSR will be a key step in furthering the cause.

L.E.A.D will set up inbound channels to answer and guide interested parties. Women should be able to reach out, enquire, and have their issues addressed to help make informed decisions.

Messaging around financial independence and security will need to be communicated to women and their families. By raising awareness around financial planning and the implications of women contributing to the household, we could see a shift in attitude towards advancing

women's participation in the workforce.

Finally, legal awareness is necessary for all stakeholders to ensure barriers or resistance to female advancement in education and employment will be seriously dealt with and punishable by law.

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